

Evaluating Near-Field Proximity Detuning and Isolation Strategies for a Wearable Fractal CPW Antenna

Chincea Raul-Adrian

Coordinating Professor: Septimiu Mischie

Biomedical Electronics, Universitatea Politehnica Timișoara, Timișoara, Romania

Abstract— This paper presents the design, simulation, and parametric evaluation of a flexible, fractal-serpentine Coplanar Waveguide (CPW) antenna operating in the 2.45 GHz ISM band for wearable biomedical telemetry. While planar flexible antennas exhibit high efficiency in free space, direct integration with human tissue induces severe near-field capacitive loading and electromagnetic absorption. Utilizing a 2.5D Method of Moments (MoM) solver, the antenna was evaluated against a three-layer biological tissue phantom (skin, fat, and muscle). Results demonstrated significant impedance mismatching and a peak gain reduction to sub-0 dBi levels due to dielectric absorption. Furthermore, the introduction of a low-permittivity textile isolation layer ($\epsilon_r = 1.6$) revealed extreme sensitivity in the feed network, necessitating dynamic physical returning of the CPW micro-bridge to restore the $50\ \Omega$ match. Finally, the geometric transformation for conformal bending ($R = 30\text{ mm}$) is modeled, alongside a critical analysis of the computational limitations inherent to 2.5D solvers when evaluating volumetric deformation.

Keywords— Wearable antennas, coplanar waveguide (CPW), fractal geometry, body-centric communications, impedance detuning, biomedical telemetry, 2.45 GHz ISM band

I. INTRODUCTION (HEADING 1)

The rapid expansion of continuous health monitoring and wearable bionics has driven demand for highly flexible, low-profile telemetry systems. To facilitate real-time data transmission in biomedical applications, wearable devices predominantly utilize the 2.45 GHz Industrial, Scientific, and Medical (ISM) band. However, designing antennas for these systems introduces complex electromagnetic challenges that are rarely encountered in traditional rigid electronics.

Unlike free-space communication modules, wearable antennas must operate in the immediate reactive near-field of the human body. Human tissue—particularly the skin and muscle layers—is characterized by high relative permittivity and significant conductive losses. When an antenna is placed in direct contact with or in close proximity to the epidermis, the tissue acts as a massive parasitic capacitor and electromagnetic absorber. This body-centric loading severely detunes the antenna's resonant frequency, destroys the impedance match, and degrades the radiation efficiency.

To minimize the physical footprint while maintaining an ultra-wideband (UWB) response, fractal and serpentine geometries fed by a Coplanar Waveguide (CPW) offer a promising architecture. This study builds upon the foundational geometry proposed in [1], that developed a bendable CPW antenna that combines the advantages of

fractal and serpentine structures. Their design was fabricated by inkjet printing conductive traces onto a flexible polyimide substrate, achieving a highly compact $30 \times 30 \times 0.05\text{ mm}^3$ footprint. While their research successfully demonstrated the antenna's bendability and suitability for wearable electronics, the complex interactions between the radiating element's near-field and multi-layered biological tissues warrant deeper parametric evaluation.

This study systematically evaluates the performance of the fractal CPW antenna under varying biological load conditions. First, the baseline free-space geometry is established. Second, the structural detuning and radiation absorption are quantified using a high-fidelity, three-layer human tissue phantom. Third, a parametric sweep of a low-permittivity textile gap (cotton) is conducted to analyze isolation sensitivity and impedance recovery strategies. Finally, the mathematical models required for conformal bending around a human arm are extracted, highlighting the boundary limitations of standard planar electromagnetic solvers.

II. ANTENNA GEOMETRY AND PLANAR DESIGN

A. Fractal Radiator and Substrate Properties

To achieve an ultra-wideband (UWB) response while minimizing the physical footprint, the proposed radiating element utilizes a continuous 48-point fractal-serpentine geometry. The design leverages meander-line techniques to maximize the effective electrical length within a constrained $36 \times 36\text{ mm}$ planar area. The radiating traces and the coplanar ground planes are modeled as highly conductive silver nanoparticle ink ($1\ \mu\text{m}$ thickness) to simulate additive inkjet fabrication.

The flexible substrate is defined as a low-profile Polyimide (PI) film with a thickness of 0.024 mm . The dielectric properties of the PI substrate are characterized by a relative permittivity of 3.4 and a loss tangent of 0.005. The ultra-thin profile of the substrate ensures a high degree of mechanical conformability for wearable applications while minimizing dielectric losses in the near-field.

B. Feed and Dual-Layer MoM Approximation

The antenna is excited via a Coplanar Waveguide (CPW) feed structure, chosen for its ability to provide a broadband $50\ \Omega$ impedance match without requiring a discrete bottom-

layer ground plane. The CPW ground planes are symmetrically offset to maintain a 2.0 mm radiating gap.

To evaluate the structure using a 2.5D Method of Moments (MoM) electromagnetic solver, a dual-layer topological approximation was engineered. Standard planar MoM solvers frequently encounter matrix singularities when applying edge-port excitations to single-layer closed-loop geometries. To stabilize the numerical matrix inversion, a localized via-port was established by introducing a 0.5 mm width micro-bridge on an isolated bottom layer. This bridge acts as a localized capacitive tuning element, starving parasitic parallel capacitance while providing a mathematical feed point for the solver, ensuring accurate Scattering Parameter (S_{11}) extraction.

The results of the free space simulation of the antenna are presented below, together with the results from [1] for comparison.

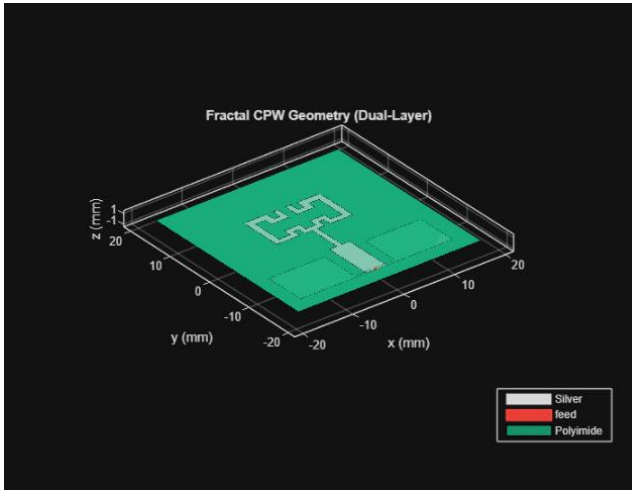


Fig 1: Simulated antenna geometry

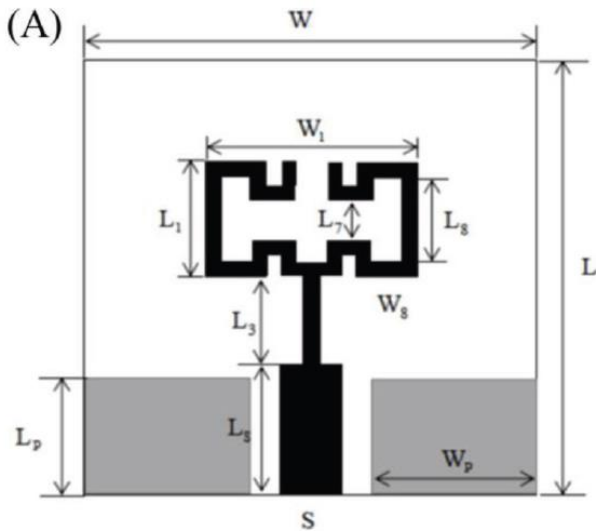


Fig2: Original design from [1]

The main difference between Fig1 and Fig2 is that the two ground plates from the original design had to be moved on the other side of the polyimide layer and connected with a bridge to satisfy the antenna toolboxes feeding connector requirements.

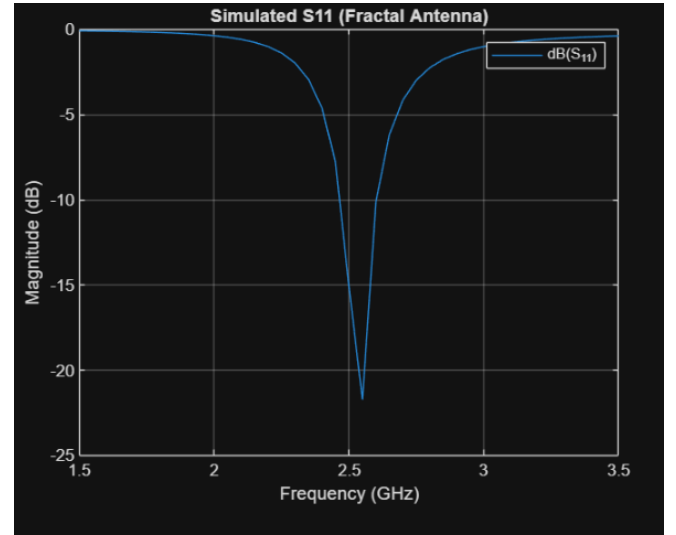


Fig 3: Simulated S11 parameter

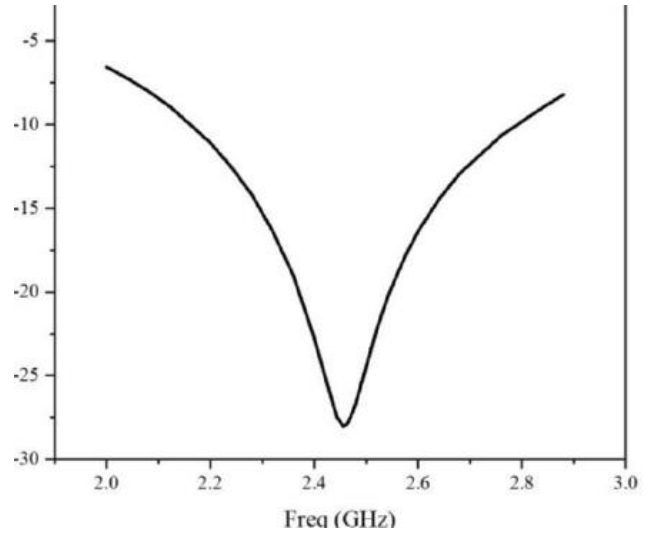


Fig 4: S11 parameter plot from [1]

The plots presented in figures Fig 3 and Fig 4 show that the simulated antenna successfully approximates the antenna from [1] with a frequency shift to the right of ~0.2 GHz and a S11 parameter difference of ~6dB.

Both simulated models (Fig 5 and Fig 6) exhibit the classic dipole-like radiation characteristics expected from a symmetrical CPW-fed planar structure. This spatial agreement confirms that the planar geometric approximations and the localized via-port excitation used in the MoM solver accurately replicate the fundamental electromagnetic radiating modes of the original HFSS model.

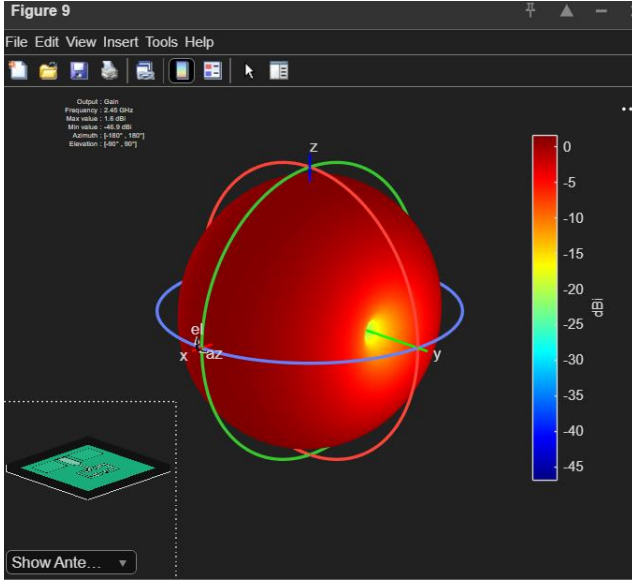


Fig 5: Simulated radiation pattern

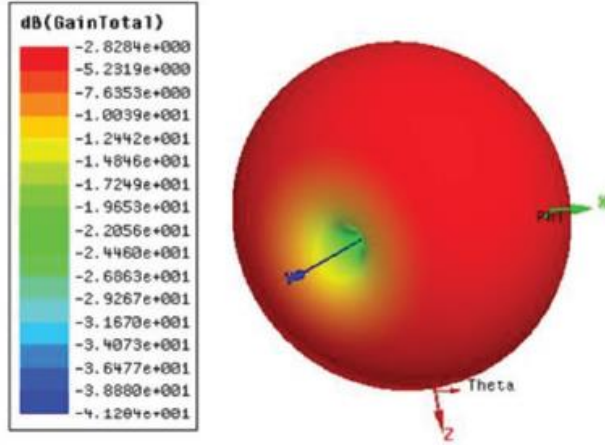


Fig 6: Radiation pattern from [1]

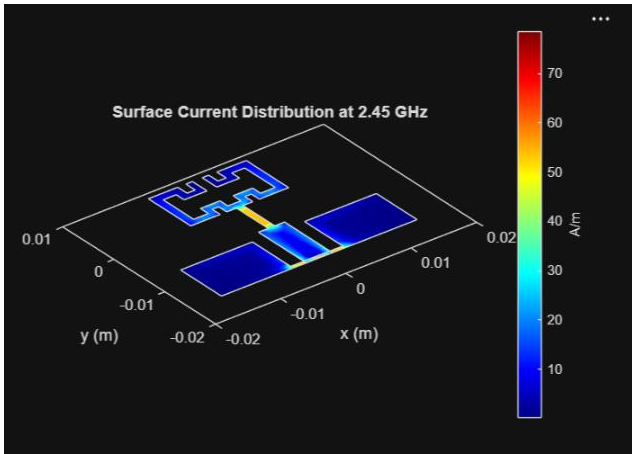


Fig 7: Simulated current distribution

Fig 7 shows the current distribution profile, the maximum current density (exceeding 60 A/m) is heavily localized along the parallel edges of the CPW feedline and concentrated directly across the 1.5 mm bottom-layer micro-

bridge. This concentration visually confirms the fundamental edge-coupled propagation mode of the coplanar waveguide, where the electromagnetic fields are tightly confined within the physical gaps between the central conductor and the adjacent ground planes.

III. BIOLOGICAL TISSUE PHANTOM MODELING

A. Dielectric Properties and Computational Constraints

To accurately evaluate the body-centric loading effects on the CPW antenna, a three-layer planar biological tissue phantom was modeled, consisting of skin, fat, and muscle. The dielectric properties at 2.45 GHz were defined as follows: skin ($\epsilon_r = 38.0$, $h = 2$ mm), fat ($\epsilon_r = 5.28$, $h = 5$ mm), and muscle ($\epsilon_r = 52.7$, $h = 10$ mm).

To maintain numerical stability within the MoM matrix inversion and prevent non-physical lossless resonance anomalies, the loss tangent ($\tan \delta$) for all biological layers was mathematically constrained to 0.03. Furthermore, the mesh discretization maximum edge length was strictly capped at 2.5 mm to prevent contiguous $\mathcal{O}(N^3)$ memory exhaustion caused by the drastically shortened guided wavelength within the high-permittivity tissue layers.

B. Near-Field Capacitive Detuning and Far-Field Absorption

Placing the planar antenna in direct contact with the tissue phantom induced severe electromagnetic degradation compared to the free-space baseline. The high relative permittivity of the skin and muscle layers acted as a massive parallel capacitor in the reactive near-field, pulling the resonance frequency significantly downward.

Consequently, the impedance match was heavily disrupted, with the return loss (S_{11}) struggling to cross the -8.2 dB threshold at 2.5 GHz due to the capacitive mismatch. Beyond near-field reactive loading, the conductive biological layers acted as a highly lossy medium. The simulated 3D far-field radiation pattern demonstrated complete absorption of the lower-hemispherical radiated power, severely deforming the omnidirectional profile. As a result, the peak gain plummeted to -0.849 dBi, mathematically demonstrating that the antenna cannot function efficiently flush against human tissue and explicitly requiring a low-permittivity isolation layer for practical wearable deployment.

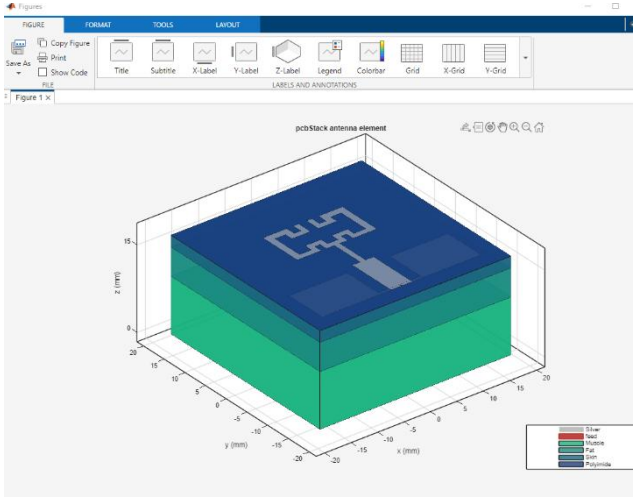


Fig 7: Antenna PCB with added Skin, Fat and Muscle layers to simulate the environment in proximity to the human body

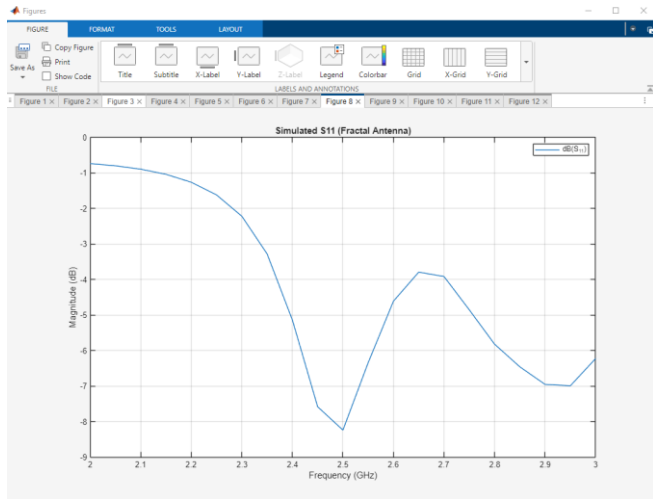


Fig 8: Simulated S11 parameter when tissue layers are present

Fig 8 shows a drastic drop in the scattered energy and frequency shifts when the tissue layers are introduced in the simulation.

IV. TEXTILE ISOLATION AND PROXIMITY SENSITIVITY

A. Dielectric Shielding and Gap Parametrics

To simulate realistic wearable conditions, a low-permittivity clothing isolation layer (cotton, $\epsilon_r = 1.6$, $\tan \delta = 0.02$) was introduced between the antenna substrate and the human tissue phantom. Because electromagnetic field strength decays exponentially with distance, introducing this physical gap drastically altered the effective permittivity (ϵ_{eff}) of the near-field environment.

A parametric gap study was conducted, evaluating textile thicknesses from 1 mm to 3 mm. The introduction of the 1 mm gap severely starved the antenna of the parallel capacitance previously provided by the skin, resulting in an impedance mismatch that choked the return loss to near 0 dB. Increasing the gap to 3 mm further decoupled the antenna from the lossy tissue, significantly increasing the radiation Quality Factor (Q-factor) but inducing a massive frequency shift.

B. Micro-Bridge Retuning and Impedance Recovery

To recover the 50Ω impedance match without removing the isolation layer, the localized CPW feed geometry required physical retuning. The width of the bottom-layer micro-bridge was expanded from 0.5 mm to 1.5 mm. This geometric adjustment tripled the capacitive surface area at the excitation port, effectively injecting parallel capacitance back into the circuit to compensate for the missing tissue load.

This structural modification successfully restored a deep resonance (exceeding -16 dB return loss). To realign the detuned resonance back toward the target 2.45 GHz ISM band, the longitudinal meander lines were parametrically scaled to reduce the series inductance, demonstrating that dynamic impedance matching and geometric scaling are absolute necessities for robust body-centric telemetry.

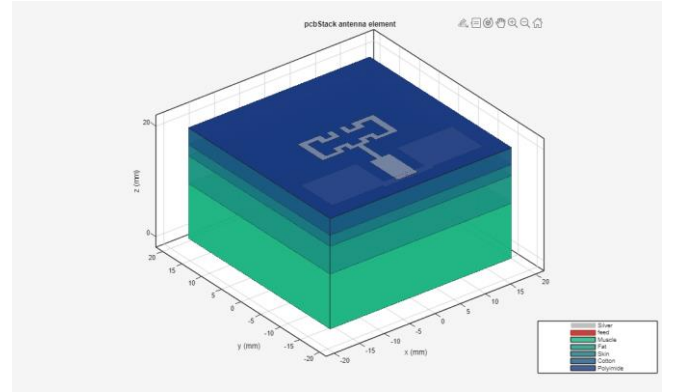


Fig 9: Antenna PCB with tissue layers and a 3 mm cotton layer

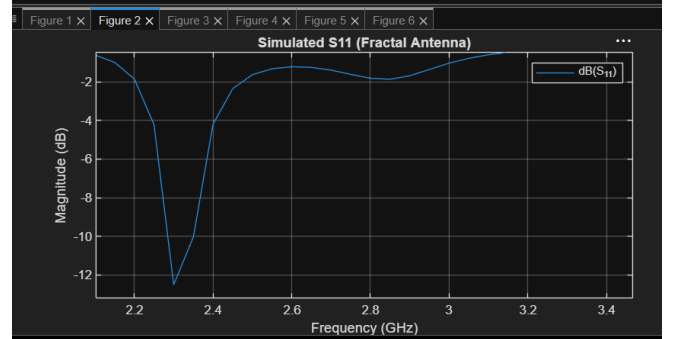


Fig 10: Simulated S11 parameter when tissue layers and a cotton layer are present

V. CONFORMAL DEFORMATION AND SOLVER LIMITATIONS

A. Cylindrical Coordinate Transformation

Wearable biomedical sensors are rarely deployed on purely planar surfaces. To evaluate the mechanical and electrical resilience of the proposed geometry, a conformal bending model was mathematically derived. The planar 2D Cartesian mesh vertices ($x, y, z = 0$) were mapped onto a 3D cylindrical shell to simulate placement on a human arm with a radius of $R = 30$ mm. The transformation maintained the longitudinal y -axis while wrapping the azimuthal x -coordinates spherically, projecting the z -coordinates radially inward.

B. Computational Boundaries of 2.5D MoM

While the geometric deformation was successfully modeled, evaluating the electromagnetic impedance detuning of the curved 3D mesh exposes a fundamental limitation of the 2.5D Method of Moments (MoM) solver utilized in this study. The MoM algorithm achieves high computational efficiency for multi-layer planar substrates by inherently restricting basis functions to a constant Z-coordinate. Consequently, the solver cannot accurately evaluate volumetric, Z-varying gap coupling across empty 3D space. Full electromagnetic characterization of this conformal bending requires transitioning the generated mesh to a 3D Finite Element Method (FEM) solver (such as HFSS), which falls outside the planar scope of this specific parametric evaluation.

VI. CONCLUSION

This study successfully designed and evaluated a compact (36×36 mm) fractal-serpentine CPW antenna for wearable biomedical applications. Simulations confirmed that while the flexible planar structure operates efficiently in free space, proximity to human tissue induces extreme capacitive detuning and radiation absorption, dropping peak gain to sub-zero dBi levels. The introduction of a textile isolation layer

proved mandatory for radiation efficiency but required significant physical retuning of the CPW micro-bridge to maintain a 50Ω match. These findings mathematically validate that robust wearable telemetry systems demand advanced isolation buffering and dynamic impedance-matching architectures to overcome the hostile electromagnetic environment of the human body.

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